ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 3RD YEAR

ARTIFICIAL INTELLIGENCE

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BSc Computer Science, 3rd year, semester 2 final examination: intelligent agents, uninformed and informed search, adversarial search, logic and probabilistic reasoning, and machine learning.

Written by Academic Central for practice, following third-year Computer Science syllabuses at South African universities (artificial intelligence). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SIX questions. Answer ALL the questions.
2. Show ALL your working. Round answers to THREE decimal places where needed.
3. $\log_2$ values: $\log_2 0.25 = -2$, $\log_2 0.75 = -0.415$, $\log_2 0.375 = -1.415$, $\log_2 0.625 = -0.678$, $\log_2 0.4 = -1.322$, $\log_2 0.6 = -0.737$.
4. Non-programmable calculators may be used.
5. Number your answers exactly as the questions are numbered.

SECTION A

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 The Turing test checks whether a machine can …

A beat a human at chess.
B hold a conversation that a judge cannot tell apart from a human's.
C learn from data.
D recognise faces.

1.2 Which search strategy is complete, and optimal when every step costs the same?

A Depth-first search
B Breadth-first search
C Greedy best-first search
D Hill climbing

1.3 A heuristic h(n) is admissible if it …

A never overestimates the true cost to reach the goal.
B always overestimates the cost.
C is always zero.
D is larger than the path cost so far.

1.4 Compared with plain minimax, alpha–beta pruning …

A can change the chosen move.
B gives the same result but examines fewer nodes.
C only works for games of chance.
D needs a heuristic that is admissible.

1.5 Which of these is an example of supervised learning?

A Grouping customers into clusters
B Classifying emails as spam using labelled examples
C A robot learning to walk from rewards
D Reducing the number of features

1.6 k-means is a …

A supervised classification method.
B unsupervised clustering method.
C reinforcement learning method.
D search algorithm.

1.7 A model that is overfitted …

A does well on the training data but badly on new data.
B does badly on all data.
C does well on all data.
D trains too slowly.

1.8 Hill climbing can get stuck at a …

A global maximum.
B local maximum.
C goal state.
D leaf node.

1.9 A single-layer perceptron cannot learn the Boolean function …

A AND
B OR
C XOR
D NOT

1.10 In reinforcement learning the agent learns mainly from …

A labelled examples.
B rewards and penalties.
C a heuristic table.
D a set of rules.

[10]

SECTION B

QUESTION 2: INTELLIGENT AGENTS

2.1 Define a rational agent. (2)

2.2 Give a PEAS description (Performance measure, Environment, Actuators, Sensors) for a self-driving minibus taxi in Johannesburg. (4)

2.3 Classify the task environment of the self-driving taxi on each of these six dimensions: observable, deterministic, episodic, static, discrete and number of agents. (3)

2.4 Explain the difference between (a) a simple reflex agent and a model-based reflex agent, and (b) a goal-based agent and a utility-based agent. (3)

[12]

QUESTION 3: SEARCH

FIGURE 1 shows a road map. The numbers on the roads are distances (path costs). The table gives a heuristic h(n), the straight-line distance from each town to the goal G. The start is S.

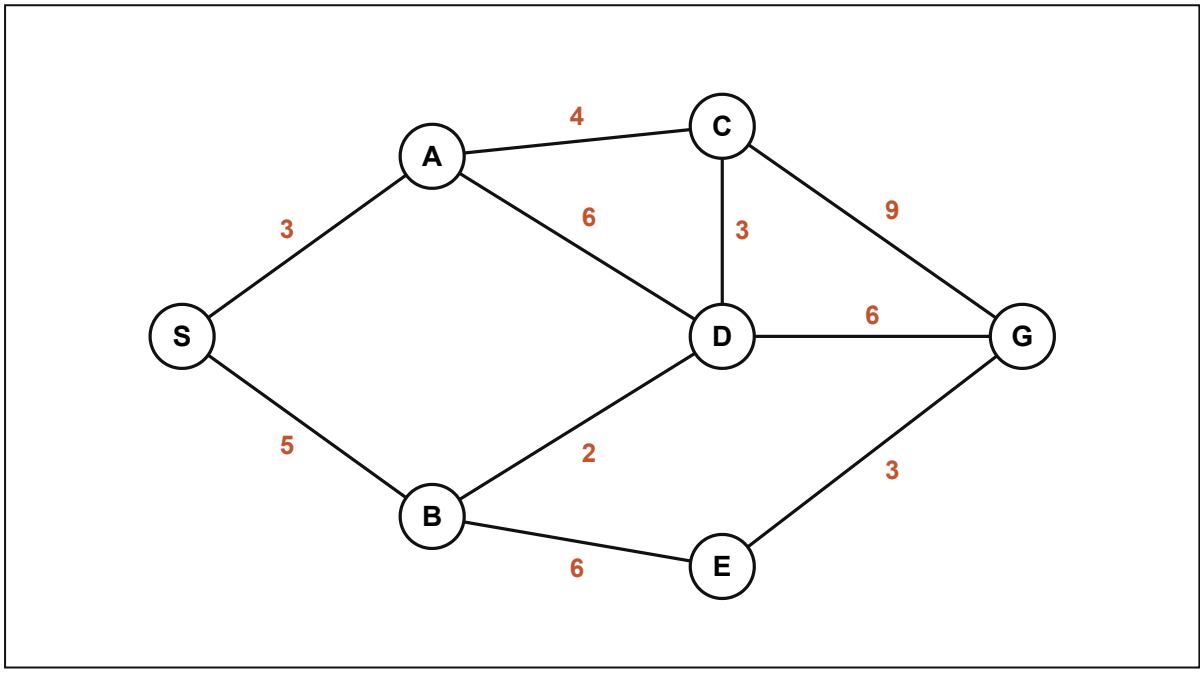


FIGURE 1: Road map from S to G

n	S	A	B	C	D	E	G
h(n)	10	9	7	8	5	3	0

Rules for 3.2, 3.4, 3.5 and 3.6: use graph search (never add a town that has already been expanded), add neighbours in alphabetical order, break ties alphabetically, and apply the goal test when a town is selected for expansion.

3.1 Formulate this as a search problem: give the initial state, the actions, the goal test and the path cost. (3)

3.2 Apply breadth-first search. Give the order in which the towns are expanded, the path found and its cost. (4)

3.3 Apply depth-first search, where the search always moves to the alphabetically first neighbour that has not been visited yet and backtracks when it is stuck. Give the order in which the towns are visited, the path found and its cost. (3)

3.4 Apply uniform-cost search. Show the frontier, with the path cost g(n) of each town, after each expansion. Give the path found and its cost. (5)

3.5 Apply greedy best-first search using h(n). Give the order of expansion, the path found and its cost. (3)

3.6 Apply A* search. Show f(n) = g(n) + h(n) for the towns in the frontier after each expansion. Give the order of expansion, the path found and its cost. (5)

3.7 Is h admissible? Explain why admissibility matters for A*. (2)

[25]

QUESTION 4: ADVERSARIAL SEARCH

FIGURE 2 shows a game tree. MAX moves first at the root A; B, C and D are MIN nodes. The numbers are the values of the end positions for MAX.

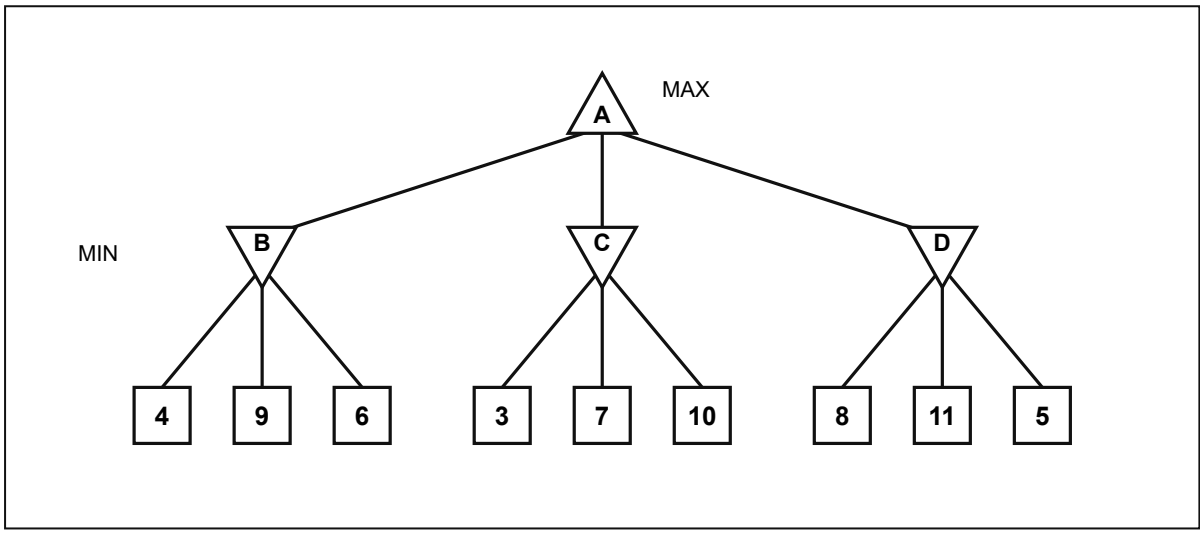


FIGURE 2: Game tree (▲ MAX node, ▼ MIN node)

4.1 Use minimax to find the value of nodes B, C, D and A. Which move should MAX make? (4)

4.2 Apply alpha–beta pruning, looking at the children from left to right. List the leaf values that are NOT evaluated, and explain why each cut is allowed. (5)

4.3 Explain why a chess program needs a depth limit and an evaluation function. (2)

4.4 Explain how expectiminimax differs from minimax, and name a kind of game it is used for. (2)

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QUESTION 5: REASONING

5.1 A knowledge base contains the sentences P ⇒ Q, Q ⇒ R and P. Convert them to clause (CNF) form and use resolution to prove R. (5)

5.2 Translate each sentence into first-order logic. Use the predicates Student(x), Studies(x), Passes(x), Lecturer(x), Robot(x) and Likes(x, y).

5.2.1 Every student who studies passes. (1)

5.2.2 Some lecturer likes every student. (1)

5.2.3 No robot is a student. (1)

5.3 2% of the emails that reach a university are phishing emails. A filter flags 95% of phishing emails, but it also wrongly flags 4% of genuine emails. Use Bayes' rule to calculate the probability that an email that has been flagged really is a phishing email. (5)

5.4 Explain why the naive Bayes classifier is called "naive". (2)

[15]

QUESTION 6: MACHINE LEARNING

The table shows data about eight students. A decision tree must predict whether a student passes.

Student	Studied	Attended	Passed
1	Yes	High	Yes
2	Yes	High	Yes
3	Yes	Low	Yes
4	No	Low	No
5	No	High	Yes
6	No	High	Yes
7	No	Low	No
8	No	Low	No

6.1 Build the decision tree with the ID3 algorithm.

6.1.1 Calculate the entropy of the whole data set. (2)

6.1.2 Calculate the information gain of Studied and of Attended. Which attribute is chosen for the root? (6)

6.1.3 Draw the complete decision tree. (2)

6.2 Six labelled points are P1 (1, 2) A, P2 (2, 3) A, P3 (3, 1) A, P4 (6, 5) B, P5 (7, 7) B and P6 (5, 6) B. A new point X is at (4, 4).

6.2.1 Calculate the Euclidean distance from X to each of the six points. (2)

6.2.2 Classify X with k-nearest neighbours for k = 3, and again for k = 5. (3)

6.3 A perceptron has the weights $w_1 = 0.2$ and $w_2 = -0.4$ and the bias $b = 0.1$. It outputs 1 if $w_1x_1 + w_2x_2 + b \geq 0$, and 0 otherwise. The learning rate is 0.5. The training example $x = (1, 1)$ has the target output 1. Calculate the output, the error, and the new weights and bias after one update with the perceptron learning rule. (4)

6.4 A spam classifier is tested on 200 emails: 40 spam emails are correctly flagged (true positives), 10 spam emails are missed (false negatives), 20 genuine emails are wrongly flagged (false positives) and 130 genuine emails pass (true negatives). Calculate the accuracy, the precision and the recall. (3)

6.5 Explain what overfitting is, and give TWO ways to reduce it. (3)

[25]

TOTAL: 100

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END OF QUESTION PAPER